Better Movement Better Life

MEDICOS MINDS

Study Smart Move Better Live Healthy

THE SCIENCE OF MOVEMENT

ORTHOPAEDICS

BONES | JOINTS | SPINE | MUSCLES | TRAUMA

Understand. Stabilize. Restore.













"Orthopaedics is not just about bones, it's about restoring dreams in motion."

KEY TOPICS

- Fractures & Dislocations
- Joint Injuries
- Spine Disorders
- Arthritis (OA/RA)
- Amputation
- Sports Injuries
- Rehabilitation
- Tumors & Infections
- Pediatric Orthopaedics
- Orthopaedic Implants & Instruments



FRACTURE FIXATION
JOINT REPLACEMENT
SPINAL SURGERY
SPORTS REHAB
FUNCTIONAL RECOVERY

Strong Bones Build Stronger Futures ♡

CONCEPT BASED | EASY TO UNDERSTAND | EXAM FOCUSED | CLINICALLY RELEVANT | HIGH YIELD NOTES

Learn Today Treat Tomorrow

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FOR MBBS | BHMS | BSC. NURSING | BDS | BPT

ORTHOPAEDICS — MEDICOS MINDS

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HIGH-YIELD BPT STUDY GUIDE & EXAM COMPANION

Orthopedics

Author: Dr. MD Kamran Alam | Instagram: @restore_healthphysio

OFFICIAL DIGITAL STUDY GUIDE VERIFICATION

Format: High-Yield Digital PDF | Standard Price: Rs. 99

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STUDY GUIDE DETAILS & CONTENTS:

Comprehensive high-yield notes designed for Bachelor of Physiotherapy (BPT), MBBS, BDS, and allied healthcare students. Covers fundamental anatomy, clinical assessments, pathological conditions, evidence-based physical therapy rehabilitation protocols, and exam-oriented high-yield review.

KEY HIGHLIGHTS:

- Exam-Focused: Concise points curated from leading medical textbooks and clinical references.
- Visual Diagrams: High-yield diagrams and tables for quick revision.
- Mobile & Tablet Optimized: Readable on phones, tablets, iPads, and laptops.
- Official Medicos Minds Release: Verified by Dr. MD Kamran Alam.

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ORTHOPAEDICS

TOPIC 4

OSTEOTOMY, BONE GRAFTING & SPINAL STABILIZATION SURGERIES

Page No. 04

1. OSTEOTOMY

Osteotomy is a surgical procedure in which bone is cut and reshaped to correct deformity, relieve pain, restore function and improve alignment.

INDICATIONS
- ✓ Angular deformities
- ✓ Limb length discrepancy
- ✓ Osteoarthritis (joint unloading)
- ✓ Congenital Deformities
- ✓ Malunion of fractures
- ✓ Non-union with deformity

TYPES
- Opening wedge osteotomy
- Closing wedge osteotomy
- Derotational osteotomy
- Corrective osteotomy
- Joint preserving osteotomy

PRINCIPLES
- ☆ Pre-operative planning – Deformity analysis, proper imaging and templating.
- ☆ Accurate bone cuts and correction.
- ☆ Stable fixation – Plates, screws, or external fixators.
- ☆ Maintain blood supply and soft tissue integrity.
- ☆ Early mobilization and rehabilitation.

TYPES OF OSTEOTOMY (EXAMPLES)



Opening wedge | Closing wedge | Derotational | Corrective

2. BONE GRAFTING

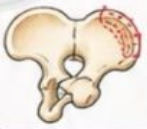
Bone grafting is the process of transplanting bone or bone substitutes to promote healing in defects, non-union or to provide structural support.

INDICATIONS
- ✓ Non-union of fractures
- ✓ Bone defects (trauma, tumor)
- ✓ Spinal fusion
- ✓ Congenital pseudoarthrosis
- ✓ Revision surgeries
- ✓ Joint fusion procedures

TYPES OF BONE GRAFT
- Autograft – From same patient (Iliac crest, fibula, tibia)
- Allograft – From another human donor (cadaver bone)
- Xenograft – From another species (bovine, coral)
- Alloplastic – Synthetic bone substitutes (HA, TCP, BMPs)

PRINCIPLES OF BONE GRAFTING
- ☆ Proper preparation of graft bed.
- ☆ Use suitable type of graft.
- ☆ Ensure close contact between graft and host bone.
- ☆ Provide stable fixation.
- ☆ Maintain blood supply.
- ☆ Prevent infection.
- ☆ Protect graft site from stress until healing occurs.

EXAMPLES



Iliac crest autograft (common site)



Cancellous bone graft (used in bone defects)

3. SPINAL STABILIZATION SURGERIES

Spinal stabilization aims to restore spinal alignment, provide stability, decompress neural elements and allow fusion.

INDICATIONS
- ✓ Spinal fractures / Dislocations
- ✓ Degenerative spinal disorders
- ✓ Spondylolisthesis
- ✓ Spinal instability
- ✓ Tumors / Infections (after decompression)
- ✓ Failed previous fusion

TYPES OF STABILIZATION
- Anterior stabilization (e.g. Vertebral body screws, Plates)
- Posterior stabilization (e.g. Pedicle screw & rod system)
- Combined (anterior + posterior)
- Minimally invasive stabilization

PRINCIPLES
- ☆ Achieve spinal stability in all three columns.
- ☆ Decompress neural elements if required.
- ☆ Restore normal alignment and balance.
- ☆ Promote solid fusion.
- ☆ Preserve as much normal anatomy as possible.
- ☆ Early mobilization with appropriate support.

COMMON INSTRUMENTATION



Pedicle screw & rod | Anterior plate | Interbody cage | Cross links & connectors

COMPLICATIONS
- ✓ Infection
- ✓ Neurological injury
- ✓ Implant failure
- ✓ Non-union (pseudoarthrosis)
- ✓ Hardware loosening
- ✓ Adjacent segment disease
- ✓ Bleeding / Dural tear



REMEMBER : Correct diagnosis → Proper planning → Right technique → Good fixation → Early rehabilitation = Successful outcome

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THANK YOU FOR YOUR PURCHASE!

We wish you the very best in your medical and physiotherapy examinations.

For any questions, supplementary questions, or study support, reach out to us:

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- Instagram: @restore_healthphysio (Daily Clinical Case Studies & Quizzes)
- WhatsApp Helpline: +91 83407 49923
- Email Support: restorehealthphysio@gmail.com

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